Employee ID Management System
Using Binary Search Tree (BST) in C++

2. Problem Statement

Managing employee identification numbers is an important task in an organization. As the number of employees increases, storing and searching employee IDs efficiently becomes necessary.

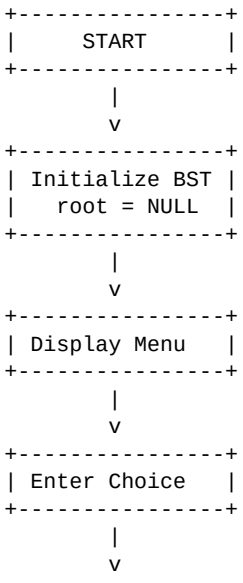
The aim of this project is to develop an Employee ID Management System using a Binary Search Tree (BST) in C++. The system allows users to insert new employee IDs, search for existing IDs, delete unwanted IDs, and display all employee IDs in ascending order.

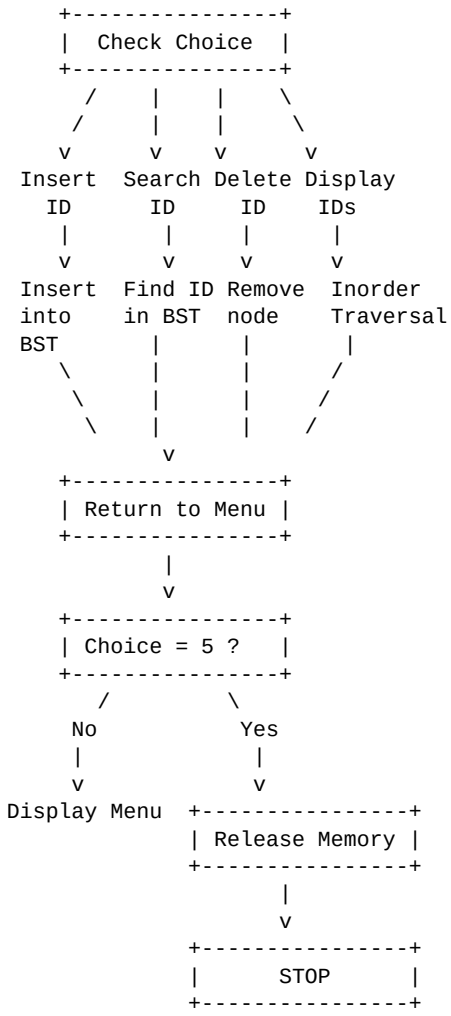
The BST organizes employee IDs so that smaller IDs are stored in the left subtree and larger IDs are stored in the right subtree. This helps perform employee ID management operations efficiently.

3. Objectives

1. To understand and implement the Binary Search Tree data structure using C++.
2. To insert employee IDs into the BST.
3. To search for a specific employee ID.
4. To delete an employee ID from the BST.
5. To display employee IDs in ascending order using inorder traversal.
6. To avoid duplicate employee IDs.
7. To understand the use of pointers, recursion, and dynamic memory allocation.
8. To develop a simple, menu-driven employee ID management application.

4. Flowchart





Flowchart explanation: The program starts by initializing an empty BST and displaying the menu. The user selects an operation to insert, search, delete, or display employee IDs. After completing the selected operation, the program returns to the menu. When the user selects Exit (option 5), the program releases allocated memory and terminates.

5. Conclusion

The Employee ID Management System using Binary Search Tree (BST) has been designed using C++. It supports insertion, searching, deletion, and displaying employee IDs in ascending order.

This project demonstrates the practical application of a BST in organizing and managing data efficiently. It also helps in understanding important programming concepts such as pointers, recursion, dynamic memory allocation, and tree traversal. Therefore, the project provides a simple and useful example of applying data structures to employee ID management.